

Negative Exponents Drill

Name: _____ Period: _____ Date: _____

DIRECTIONS: Simplify each expression. Write ALL answers with **POSITIVE exponents only**. Show your work!

1. x^{-5}

2. 3^{-2}

3. y^{-1}

4. 2^{-4}

5. $a^{-3} \cdot a^5$

6. $b^7 \cdot b^{-10}$

7. $\frac{1}{x^{-3}}$

8. $\frac{1}{y^{-7}}$

9. $\frac{5}{a^{-2}}$

10. $\frac{1}{2^{-5}}$

11. $\frac{3}{m^{-4}}$

12. $\frac{1}{(-2)^{-3}}$

13. $3x^{-2}$

14. $(3x)^{-2}$

15. $5a^{-4}$

16. $(5a)^{-4}$

17. $-2y^{-3}$

18. $(-2y)^{-3}$

19. $4m^{-1}$

20. $(4m)^{-1}$

21. $\frac{x^{-2}}{y^{-3}}$

22. $\frac{a^{-5}}{b^2}$

23. $\frac{m^3}{n^{-4}}$

24. $\frac{2x^{-3}}{y^{-1}}$

25. $\frac{3a^{-2}}{5b^{-4}}$

26. $\frac{x^{-1}y^2}{x^3y^{-2}}$

27. $x^{-1/2}$

28. $y^{-2/3}$

29. $8^{-1/3}$

30. $16^{-3/4}$

31. $27^{-2/3}$

32. $\frac{1}{x^{-1/4}}$

33. $\frac{1}{3x^{-2}}$

34. $\frac{1}{2a^{-3}}$

35. $\frac{1}{5y^{-1}}$

36. $\frac{4}{2m^{-4}}$

37. $\frac{1}{3x^{-2/3}}$

38. $\frac{6}{2b^{-1/2}}$

39. $(x^{-3})^2$

40. $(y^{-2})^{-4}$

41. $(2x^{-3})^{-2}$

42. $(3a^{-2}b^3)^{-1}$

43. $(x^{-2}y^{-3})^{-2}$

44. $\left(\frac{a^2b^{-1}}{c^{-3}}\right)^{-2}$

45. $(x^{-3}y^2)^{-2} \cdot (xy^{-1})^3$

46. $\frac{(2a^{-2}b)^3}{(4a^{-1}b^{-2})^2}$

47. $\left(3x^{-1/2}y^{2/3}\right)^{-6}$

48. $[(x^{-2})^3 \cdot x^5]^{-1}$

49. $\left(\frac{a^{-3}b^2}{a^2b^{-5}}\right)^{-2} \cdot (ab)^{-1}$

50. $\frac{1}{2x^{-3/2} \cdot (4x)^{-1/2}}$

x^2	$\frac{1}{y}$	$\frac{x^9}{y^7}$	$\frac{1}{625a^4}$	$3\sqrt{b}$	$\frac{y^3}{x^2}$	$\frac{1}{9}$	$\frac{1}{8}$	y^8	$\frac{1}{4m}$
x	$\frac{1}{\sqrt{x}}$	$\frac{a^2}{3b^3}$	$\frac{3}{x^2}$	$\frac{1}{a^5b^2}$	-8	$\frac{x^6}{4}$	$\frac{y}{5}$	$\frac{5}{a^4}$	$\frac{1}{x^5}$
32	$\frac{b^7}{2a^4}$	$\frac{x^3}{729y^4}$	$\frac{1}{x^6}$	$\frac{y^4}{x^4}$	$\frac{1}{9}$	a^2	$\frac{4}{m}$	$\frac{a^9}{b^{15}}$	$\frac{\sqrt[3]{x^2}}{3}$
$\frac{1}{9x^2}$	x^3	$2m^4$	$\frac{1}{16}$	m^3n^4	$\frac{3b^4}{5a^2}$	$\frac{-2}{y^3}$	$\frac{1}{b^3}$	$\frac{1}{\sqrt[3]{y^2}}$	$5a^2$
$\frac{b^2}{a^4c^6}$	x^4y^6	$3m^4$	$\frac{2y}{x^3}$	$\frac{1}{2}$	$\sqrt[4]{x}$	$\frac{a^3}{2}$	$\frac{-1}{8y^3}$	$\frac{x^2}{3}$	y^7